# 1. Clinical Question & Core Hypothesis

## Clinical Question

The Ontologist highlighted a tension between immune-contexture readouts (Immunoscore/TILs/TLS) and liquid-biopsy signals (ADNP mRNA in CTCs) in NMIBC—i.e., why do some tumors with apparently favorable local immune organization still fail therapy (BCG) or recur? 2) Tier 1 paths used: I built on Core Path 1 (ADNP mRNA detectable in CTCs as a tumor-intrinsic, liquid-biopsy signal) and Core Path 3 (immune-contexture/Immunoscore concept capturing CD3/CD8 infiltration). These provide a tumor-intrinsic vs. local-immune interaction axis that is mechanistically testable. 3) Tier 2 support: Immunotherapy-biomarker concepts (PD-L1, TMB, HPV analogies) and AI-based biomarker accessibility justify (a) that therapy response is modulated by both tumor and immune features and (b) that integrated tissue + liquid signatures can be operationalized; MSI and other genomic features are plausible modifiers of immune responsiveness. 4) Choice of endpoint: 'therapy_response' (BCG response proxy) is the most directly actionable clinical problem in NMIBC—identifying who will/won't benefit from intravesical immunotherapy was emphasized by the Ontologist and is well matched to combining tissue immune organization with circulating tumor signals. 5) Deliberate exclusions: I excluded Tier 3 exploratory concepts (CSF AD biomarkers, SARIFA, pZVR) as speculative and not directly relevant to NMIBC immune–liquid biopsy interplay; I also kept the hypothesis to only two anchor molecular features (ADNP and CD8A) to maintain mechanistic clarity?

## Working Hypothesis

In NMIBC, tumor-cell intrinsic high ADNP expression reprograms the TLS-like lymphoid microenvironment by altering the secretome and antigen presentation, leading to impaired CD8+ T cell effector function (reduced granzyme B, perforin, and IFN-γ, increased exhaustion markers) and enhanced CTC dissemination. Clinically, TLS-high/ADNP-high NMIBC tumors will exhibit reduced intravesical BCG therapy response compared with TLS-high/ADNP-low tumors, as assessed by the predefined therapy_response endpoint. This prediction can be tested by integrating tissue TLS scoring, CD8A activity measurements, and ADNP status in CTCs, alongside standard modifiers (MSI, PD-L1, TMB, AI-based biomarkers) to model response heterogeneity.

| Element | Value |
|---|---|
| Imaging feature | TLS-like lymphoid aggregates |
| Anchor gene(s) | ADNP, CD8A |
| Primary endpoint | therapy_response |

# 2. Why This Gene — Not Others

## Primary hypothesis gene(s)

**ADNP:** ADNP mRNA detectable in CTCs as a tumor-intrinsic, liquid-biopsy signal) and Core Path 3 (immune-contexture/Immunoscore concept capturing CD3/CD8 infiltration).

**CD8A:** CD8A) to maintain mechanistic clarity.

## Alternative / secondary genes considered

| Tier | Gene | Role / Reason excluded from core |
|---|---|---|

| | | |
|---|---|---|
| Secondary | MSI | Supporting modifier from knowledge graph |
| Secondary | PD-L1 | Supporting modifier from knowledge graph |
| Secondary | TMB | Supporting modifier from knowledge graph |
| Secondary | CTC count/phenotype | Supporting modifier from knowledge graph |
| Secondary | AI-based biomarker integration | Supporting modifier from knowledge graph |

## 3. Clinically Testable Predictions

**Prediction 1:** a mechanistic claim linking ADNP to CD8+ T cell impairment and CTC shedding within TLS-rich regions, and

**Prediction 2:** a clinical consequence of reduced BCG response in TLS-high/ADNP-high tumors. The approach leverages available Tier 1 signals (ADNP in CTCs, TLS/CD8A status) and Tier 2 modifiers to build an integrative biomarker framework.

**Prediction 3:** At the tissue level, tumors with the imaging feature but low anchor gene activity will show immune cells confined to structural peripheries rather than infiltrating tumor nests.

## 4. Minimum Viable Validation Plan

### A. Retrospective Cohort Analysis

**Data source:** TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

### B. Tissue-Level Validation

**Methods:** Multiplex IF / IHC / spatial transcriptomics

**Key readouts:** - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

**Goal:** Confirm that the proposed mechanism operates at the spatial / tissue level.

### C. Functional Validation

**Models:** Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

**Perturbations:** Genetic (KO/rescue) or pharmacologic modulation of anchor gene

**Readouts:** T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

**Goal:** Establish causal relationship between anchor gene activity and immune efficacy.

# 5. Strengths, Caveats & Alternative Directions

## Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

## Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

## Alternative Directions

- MSI: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- PD-L1: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.